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RESEARCH ARTICLE

Optimizing CNN-GRU Hybrid Ratios for Resource-Constrained Audio Classification: A Systematic Study From Parameter Efficiency to MCU Deployment

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ABSTRACT Urban sound classification has become a critical enabling technology for Internet of Things (IoT) applications, smart cities, and environmental monitoring systems. Despite advances in deep learning, deploying these models on resource-constrained microcontroller units (MCUs) poses significant challenges in achieving a balance between classification accuracy, computational efficiency, and energy consumption. This study introduces the first comprehensive and systematic analysis of hybrid CNN-GRU architectures optimized for embedded audio processing. Thirteen distinct layer ratio configurations were evaluated under varying temporal chunk sizes across three representative MCU platforms. Experiments conducted on the UrbanSound8K dataset reveal that CNN-heavy configurations achieve optimal performance for ultra-short duration processing, reaching 93.92% accuracy at 0.0625 seconds, while balanced architectures deliver the best trade-offs for medium-duration segments. On the ESC-50 dataset involving two different chunks, the experiments reveal mostly similar performance trends. With both datasets, the performance of the models peaks at about 3-seconds chunks. The study identifies eight critical application scenarios across safety, healthcare, industrial, and smart environment domains, establishing optimized architecture-platform combinations for each use case. Deployment results demonstrate that CNN-heavy models achieve more than 93% accuracy with sub-20ms inference times for ultra-short segments, enabling real-time tasks such as emergency vehicle and glass break detection. The proposed framework provides a quantitative foundation for selecting the most suitable architecture–platform pairings under specific application constraints, effectively bridging the gap between academic research and practical embedded audio classification systems.

INDEX TERMS Audio classification, audio chunks, CNN-GRU hybrid networks, embedded systems, feature maps, layer ratios, microcontroller deployment, resource constraints, IoT applications.

I. INTRODUCTION

The proliferation of Internet of Things (IoT) devices and the evolution of edge computing have led to an unprecedented demand for intelligent audio processing within highly resource-constrained environments. Audio classification now plays a vital role in smart cities, industrial monitoring, healthcare, and autonomous systems, driven by growing needs for privacy-preserving computation, reduced cloud dependency, and real-time responsiveness [1], [2]. However, implementing deep learning models on microcontroller units (MCUs) introduces critical challenges related to limited computational capacity, memory, and power budgets. Achieving a balance between accuracy and efficiency within such constrained hardware remains an active area of research.

Recent developments in edge-based audio classification have demonstrated significant progress. Compressed neural networks have achieved reductions of up to 97% reductions in model size while maintaining near state-of-the-art accuracy on established benchmarks [3]. State-of-the-art models currently attain between 78% and 96% accuracy on datasets such as UrbanSound8K and ESC-50, with certain optimized

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architectures surpassing 97% accuracy on ESC-50 [4], [5]. Nonetheless, deploying these models on microcontrollers remains complex due to stringent resource limits; typical MCUs offer only 32 KB to 2 MB of flash memory, 128 KB to 1.4 MB of RAM, and operate within computational budgets of 20-100 MIPS, often consuming less than 100 mW for battery-powered devices [6], [7].

Hybrid CNN-RNN architectures have emerged as particularly promising for audio classification tasks, as they combine the spatial feature extraction strength of convolutional neural networks with the temporal sequence modelling of recurrent units. CNN-GRU hybrid architectures consistently outperform single-architecture approaches, offering 5-10 % accuracy improvements while reducing parameters by 25-40 % compared to CNN-LSTM equivalents [8], [9]. Empirical studies have shown that CNN-BiGRU architectures achieve accuracies as high as 89.30 %, compared to 76.40 % for CNN-LSTM models on music classification tasks [10]. Moreover, GRU-based hybrids offer 25-30 % lower memory usage and train 15-30 % faster than LSTM-based designs, making them appealing for embedded deployment [11].

Although recent advances in environmental sound classification have demonstrated effective deployment of deep learning models on embedded devices, [12], [29], a critical gap remains regarding systematic optimization of CNN-GRU layer ratios for resource-constrained systems. While surveys such as Ding et al. [30] provide broad analyses of acoustic scene classification and Gairí et al. [31] review environmental sound recognition on embedded devices, little research has focused on how specific CNN-to-GRU layer ratios influence trade-offs among parameter efficiency, classification accuracy, and deployment feasibility on MCU hardware. Recent contributions in hardware-aware neural architecture search [32] and general CNN-RNN hybrid designs [33], [34] have emphasized automation and high-level design exploration but have not systematically examined manual ratio optimization. Similarly, studies that address extreme deployment constraints, such as those by Mohaimenuzzaman et al., [12], emphasize optimization pipelines rather than architectural configurations.

Another underexplored aspect is the influence of temporal chunk duration on performance of CNN-GRU within resource-constrained environments. Temporal chunking has direct impact on memory utilization, processing latency, and real-time feasibility, yet few studies have examined these dependencies in depth. The current state of research lacks a holistic framework that jointly considers architectural structure, temporal chunking strategies, and hardware deployment constraints on mature MCU platforms. Prior studies typically isolate either algorithmic design innovations or hardware-level optimizations without providing systematic guidance for practitioners developing real-world embedded systems.

To address these gaps, this work presents a comprehensive systematic analysis of CNN-GRU hybrid architectures optimized for embedded audio processing. The study focuses on mature and widely deployed MCU families such as the STM32F4 and STM32H7 series, which represent realistic deployment environments for contemporary edge audio applications. Although newer NPU-enabled solutions such as the STM32N6 demonstrate potential for specialized inference acceleration, they involve fundamentally different optimization dynamics and are therefore beyond the scope of the present study. Our investigation concentrates on software-optimized architectures that can be immediately deployed in current embedded ecosystems, where traditional MCUs already achieve 89-94% accuracy at substantially lower cost and system complexity.

The contributions of this work are threefold. First, the paper presents an extensive empirical evaluation of different CNN-GRU layer ratio configurations under multiple resource-constrained scenarios, establishing quantitative relationships between architectural choices and deployment performance metrics, including accuracy, latency, memory, and energy efficiency. Second, it systematically investigates the impact of temporal chunk duration ranging from 0.0625 to 3.0 seconds, providing clear guidelines for balancing accuracy and real-time responsiveness. Finally, it identifies eight critical use cases across various domains including safety, healthcare, industrial IoT, and smart environments, deriving optimal architecture and platform recommendations validated through extensive experimentation on UrbanSound8K and ESC-50 datasets, followed by deployment on actual MCU development boards.

II. LITERATURE REVIEW

A. CNN-GRU HYBRID ARCITECTURES FOR AUDIO CLASSIFICATION

Recent advancements in CNN-GRU hybrid architectures have significantly enhanced the performance of audio classification systems. Studies have demonstrated that these architectures effectively combine the spatial feature extraction strengths of convolutional neural networks (CNNs) with the temporal modelling capabilities of recurrent networks such as gated recurrent units (GRUs). Ashraf et al. [13] reported that CNN combined with bidirectional GRU layers achieved an accuracy of 89.30% on music classification tasks, representing a 13% improvement over traditional CNN-LSTM architectures. Similarly, Zhang et al. [14] introduced attention-based convolutional recurrent neural networks (ACRNN), incorporating frame-level attention mechanisms that yielded 93.1% accuracy on the UrbanSound8K dataset and 84.4% on ESC-50.

Comprehensive architectural analyses have consistently indicated that CNN models with approximately five convolutional layers and 128-256 filters offer optimal feature extraction efficiency, while GRU components with one to two layers and 128-512 hidden units provide effective temporal modelling [15]. Moreover, mel-spectrogram features have been found to outperform MFCC representations by approximately 13 percentage points across multiple datasets [16].

B. EMBEDDED DEEP LEARNING FOR AUDIO PROCESSING

Parallel to architectural advances, considerable progress has been made in the deployment of deep learning models within resource-constrained embedded environments. Recent developments have enabled ultra-lightweight models to operate with memory footprints as low as 15-50 kB while maintaining classification accuracy of 85-95% [7]. Comprehensive reviews by Gairí et al. [31] have systematically analysed deep learning–based sound recognition on embedded devices, while Chang et al. [35] proposed multi-level transfer learning frameworks for environmental sound classification. Mou and Milanova [36] examined model compression techniques tailored for audio classification at the edge, and İşler [37] conducted comparative performance analyses of deep learning models deployed in IoT-fog computing frameworks for urban sound recognition.

Mohaimenuzzaman et al. [12] presented a detailed deployment pipeline for environmental sound classification on severely resource-limited hardware, demonstrating the feasibility of deep acoustic networks when carefully optimized across preprocessing, architecture design, and post-processing stages. Complementary work by Goulão et al. [38] investigated effective training methodologies for real-world edge deployment, and Bellafkir et al. [39] addressed urban sound classification under tight resource constraints, further validating the growing practicality of deep learning models in embedded audio analysis.

TensorFlow Lite Micro has emerged as the dominant framework for such deployments, providing a stable ecosystem for low-power neural inference. The integration of CMSIS-NN optimization libraries has resulted in performance improvements of up to 4.6 times in runtime throughput and 4.9 times in energy efficiency on ARM Cortex-M processors. Additionally, post-training quantization techniques have achieved up to fourfold reductions in model size with minimal loss in accuracy, while quantization-aware training has preserved accuracy within a 2% margin compared to full-precision models [18].

C. MCU PLATFORM CAPABILITIES AND CONSTRAINT

Modern MCU platforms have demonstrated increasing suitability for executing complex neural networks under strict performance and power limitations. Several recent studies have documented the successful deployment of deep models with inference times ranging from 8 to 150 milliseconds, depending on architecture complexity and input dimensions [19]. Advanced deployment frameworks have reported up to 2.3 times faster inference times compared to TensorFlow Lite Micro, coupled with 20-30% reductions in memory utilization [20].

For real-time implementations, maintaining processing times below 32 milliseconds for 16 kHz audio streams has proven critical. This performance threshold has been achieved through techniques such as double-buffering, direct memory access (DMA)-based data transfers, and CMSIS-DSP, optimized feature extraction, which reduce preprocessing latency to 5-15 milliseconds [21]. Importantly, the temporal buffer size, equivalent to the chunk duration analysed in this study, plays a central role in balancing classification accuracy and real-time feasibility. The interplay between computational capability, available memory, and chunk size remains a decisive factor in determining the practical viability of MCU-based audio classification deployments.

D. RESEARCH GAP AND POSITIONING

While substantial progress has been achieved in both model design and embedded optimization, the systematic study of CNN-GRU layer ratio optimization for constrained hardware remains limited. Recent comprehensive reviews [30], [31] have surveyed environmental and acoustic sound classification broadly but did not specifically address the proportional relationship between CNN and GRU components. Similarly, works such as those by Ranmal et al. [32] on neural architecture search and Bansal and Garg [33], [34] on CNN-RNN hybrids have advanced general hybrid modelling but without offering systematic comparisons across CNN-GRU ratios under deployment constraints.

Mohaimenuzzaman et al. [12] focus on deployment pipeline optimization for extremely constrained devices but do not provide systematic architectural comparison across different CNN-GRU ratios. Most studies focus on either architectural innovations or hardware optimizations in isolation, without providing comprehensive guidance for joint optimization of architecture design and deployment constraints across diverse application scenarios.

The present study directly addresses this research gap by conducting a comprehensive comparative analysis of CNN-GRU hybrid ratios across multiple MCU platforms, varying chunk durations, and real-world application domains. This work represents the first systematic effort to establish a quantifiable relationship between architecture composition and deployment feasibility within resource-limited embedded environments, offering actionable insights for practitioners targeting mature MCU platforms that reflect the current realities of edge audio processing.

III. METHODOLOGY

A. HYBRID CNN-GRU ARCHITECTURE DESIGN

The experimental design of this study has involved developing thirteen distinct CNN-GRU hybrid architectures (Figure 1 and also Table 11 of Appendix-1) to systematically examine how different ratios between convolutional and recurrent layers affect audio classification performance and deployment efficiency. The models are organized into five groups based on their CNN-to-GRU layer ratios, with an additional specialized configuration that incorporates depth-wise separable convolutions. The first group, referred to as *GRU-heavy* architectures, emphasizes temporal modelling and consists of configurations with ratios of 1:3, 1:4, and 1:5. The second group, denoted as *balanced* configurations, maintains an equilibrium between spatial and temporal processing through ratios of 2:2, 2:3, and 2:4. The third group, categorized as *CNN-heavy* architectures, focuses on intensive spatial feature extraction, adopting 3:1, 3:2, and 3:3 ratios. The fourth group, representing *CNN-dominant* configurations, further extends convolutional processing with ratios of 4:1 and 4:2. Finally, the fifth group, labelled *CNN-maximum*, maximizes spatial feature extraction through a 5:1 configuration. The special mixed 2:2:2 model incorporates depth-wise separable convolutions to evaluate parameter efficiency while retaining representational power.

Each architecture follows a unified design pattern to ensure experimental consistency. The models consist of sequential one-dimensional convolutional layers, each followed by batch normalization and max-pooling operations for feature extraction. The recurrent block comprises GRU layers augmented by attention mechanisms to enhance temporal feature weighting. The classification stage includes fully connected dense layers with dropout regularization to prevent overfitting. Input features are extracted from 39-dimensional Mel-frequency cepstral coefficients (MFCCs) derived from pre-processed audio chunks. The spatial down-sampling ratio is tailored to each architecture to maintain comparable feature dimensionality across varying CNN depths.

The main parameters of the hybrid models, including layer configuration, parameter count, FLOPS and Inference speed are summarized in Table1. The models are designated as n_1:n_2, where n_1 and n_2 represent the number of 1D convolutional and GRU layers respectively. The last model in the table is denoted as n_1:n_2:n_3, where n_3 represents the number of depth-separable convolution layers. As shown in Table 1, the overall parameter count, FLOPS, and Inference speed change in a predictable manner as the number of 1D convolutional and GRU layers is varied. The models with increasing GRU layers exhibit slower inference speeds due to the increased temporal modeling capacity. Increasing Conv1D layers enhances feature extraction at the expense of a higher computational load. Some combinations such as 2:3, and 4:2, show high FPS, indicating an optimal trade-off between spatial feature extraction and sequence modeling. On the other hand, the depth-separable convolutions in 2:2:2 further reduce the computational overhead and maintain a competitive speed.

B. EXPERIMENTAL SETUP

The architectures are evaluated on two widely recognized benchmark datasets: UrbanSound8K and ESC-50. The UrbanSound8K dataset contains 8,732 labelled audio clips distributed across ten urban sound classes, whereas ESC-50 comprises 2,000 environmental sound recordings encompassing fifty categories [22], [23]. Together, these datasets provide a comprehensive evaluation framework that covers both urban and natural soundscapes, ensuring generalizable results across diverse environmental contexts.

Audio preprocessing follows a consistent pipeline involving waveform normalization, resampling to 16 kHz, and MFCC feature extraction with 39 coefficients per frame. Data augmentation techniques such as time-stretching (1.5×), pitch-shifting (+2 semitones), and additive noise (zero-mean Gaussian noise with amplitude sampled uniformly in the range of 0.005-0.008), are applied to enhance model robustness and generalization.

TABLE 1. **Hybrid architectures and parameters.**

Model	Parameters	FLOPS	Inference speed (FPS)
1:3	109962	41980	554.1
1:4	103562	30556	462.2
1:5	128522	30556	400.7
2:2	66250	280476	264.7
2:3	116170	280476	557.3
2:4	147466	1033340	368.7
3:1	78794	776476	466.7
3:2	103754	776476	611.4
3:3	153674	776476	435.8
4:1	128586	1270684	374.1
4:2	153546	1270684	633.0
5:1	252746	1864348	379.5
2:2:2	71114	348988	483.4

To explore the influence of temporal granularity, the study systematically analyses seven audio chunk durations: 3.0, 1.0, 0.75, 0.5, 0.25, 0.125, and 0.0625 seconds. These durations enable assessment of how contextual length affects classification accuracy, ranging from long segments containing complete environmental events to ultra-short fragments suitable for real-time embedded processing.

Three representative MCU platforms from the STM32 family are selected to evaluate the feasibility of deployment across distinct performance and resource classes. The STM32F411, featuring a 100 MHz ARM Cortex-M4 core with 512 kB of flash memory and 128 kB of SRAM, represents a cost-optimized deployment. The STM32F429, equipped with a 180 MHz Cortex-M4 core and larger memory capacity (2 MB of flash and 256 kB of SRAM), serves as the mainstream performance platform. Finally, the STM32H743, powered by a 480 MHz ARM Cortex-M7 core with 2 MB of flash and 1 MB of SRAM, represents high-performance embedded hardware optimized for real-time applications. During the evaluation, the STM32 Profiler Kit was used to measure the power consumption of the aforementioned MCUs during various phases of signal classification.

Training is conducted using 10-fold cross-validation with a 9:1 training-validation split. Early stopping is implemented to mitigate overfitting, while adaptive learning rate scheduling facilitated efficient convergence. All models employ the Adam optimizer and categorical cross-entropy loss function.

C. USE CASE IDENTIFICATION AND ANALYSIS

To ensure real-world applicability, eight representative use cases are identified and analysed across four principal domains: safety, healthcare, industrial IoT, and smart environments. Each domain reflects a unique balance of performance, latency, and power efficiency requirements.

FIGURE 1. Design schematic of hybrid model groups.

In the *safety domain,* applications such as emergency vehicle detection and glass break detection demand ultra-low latency (below 200 milliseconds) and high accuracy (above 95%) to support real-time decision-making in security and emergency systems. In the *healthcare and wellness domain*, use cases such as baby cry detection and respiratory monitoring prioritize privacy-preserving edge processing with moderate accuracy requirements of 85% to 92%, making them suitable for home healthcare and elderly monitoring. The *industrial IoT domain* encompasses predictive maintenance and motor fault diagnosis, emphasizing accuracy above 95% and processing windows from medium to long durations for machinery and production monitoring. Lastly, the *smart environment domain* includes smart home appliance monitoring and environmental sound recognition, focusing on cost and energy efficiency while maintaining accuracy above 80% for large-scale IoT deployments.

D. PERFORMANCE METRICS AND EVALUATION FRAMEWORK

The performance evaluation framework is designed to capture the multi-dimensional nature of embedded model deployment. Four primary evaluation dimensions are defined to assess architecture effectiveness comprehensively. First, classification performance is evaluated using accuracy, precision, recall, and F1-score metrics across all sound classes to ensure both global and class-specific assessment. Second, computational efficiency is characterized in terms of inference latency, memory consumption, and parameter count, quantifying the trade-offs between architectural complexity and deployment feasibility. Third, deployment feasibility is assessed by examining compatibility with MCU memory limits, real-time inference capabilities, and energy consumption under continuous operation. Finally, application suitability is determined by analysing how each architecture's performance profile aligns with the operational requirements of specific use cases. To guarantee statistical rigor, all results have undergone significance testing using 95% confidence intervals and multiple comparison corrections (Appendix). The evaluation protocol thereby ensured robust, interpretable comparisons across the extensive experimental configurations.

IV. RESULTS AND ANALYSIS

A. OVERALL ARCHITECTURE PERFORMANCE ANALYSIS

A comprehensive evaluation is conducted across all thirteen CNN–GRU hybrid architectures and seven temporal chunks, derived from UrbanSound8K (US8K) and two chunks from ESC-50, to examine how architectural configuration and temporal granularity influence classification performance. The average accuracy results for representative chunk durations are presented in Table 2 and Table 3. The results reveal distinct performance patterns that highlight the effect of architectural composition on classification accuracy. CNN-heavy architectures consistently outperform other configurations across all chunk durations. The CNN-Maximum 5:1 model demonstrates the highest overall mean accuracy of 97.9%,

TABLE 2. Model accuracy with US8K chunks.

Model (FP32)	Overall Test Accuracy (%) of US8K Chunks				
	0.0625s	0.125s	0.25s	0.5s	0.75s
5:1	93.92	96.77	98.16	98.17	98.90
4:1	92.48	96.37	97.77	97.84	98.44
4:2	92.95	95.33	96.27	97.61	98.21
3:1	90.11	93.63	95.99	96.82	97.58
3:2	91.18	93.46	95.03	96.77	97.22
3:3	87.64	93.12	94.79	97.64	95.78
2:2	89.36	92.07	95.13	96.77	95.09
2:3	89.96	93.07	96.03	97.94	97.09
2:4	87.54	91.85	95.84	96.44	97.51
1:3	88.57	91.16	93.69	95.05	97.12
1:4	87.83	94.25	94.81	97.37	95.84
1:5	91.26	94.19	93.57	97.07	96.85
2:2:2	90.19	93.11	94.27	97.70	95.54

maintaining strong performance across short and medium temporal segments. For ultra-short durations of 0.0625 s, CNN-dominant models, such as 5:1 (93.92%), 4:1 (92.48%), and 4:2 (92.95%), outperform GRU-heavy counterparts such as 1:3 (88.57%) and 1:4 (87.83%), demonstrating their robustness in scenarios with limited temporal context.

Performance also varies with chunk duration. While CNN-heavy architectures dominate short-duration tasks, GRU-heavy models exhibit superior accuracy for longer segments, with the 1:5 configuration achieving 99.58% accuracy at a 3.0 s duration (Figure 2 (a)-Figure 2(b)). This observation suggests that temporal modelling becomes increasingly beneficial when longer contextual information is available.

Balanced architectures, including 2:2, 2:3, and 3:2 ratios, deliver consistently high performance across all durations ranging from 0.25s to 1s, and achieve stable accuracy between 95% and 98%. These balanced models are therefore well-suited for applications requiring flexibility across variable audio segment lengths.

The accuracy performance of all the models, with 5s and 3s chunks derived from the ESC-50 dataset, reveals a mostly similar trend, although it is 2 to 7 percentage points lower across the models as compared to US8K chunks. For instance, with 3s chunk, the CNN-maximum 5:1 model shows an overall accuracy of 97.10%, about 2 percentage points lower than that of US8K. However, for the same 3s chunk of ESC-50, the accuracy drops by about 6 to 7 percentage points, compared to US8K, for GRU-heavy (1:5 and 1:4) and 2:2:2 models. It is interesting to note that the performance of models peaks at about 3-seconds chunks, for both datasets, indicating that for chunk durations beyond this, they mostly add noise rather than new discriminative information.

B. CHUNK DURATION IMPACT ANALYSIS

An in-depth examination of chunk duration effects reveals critical trends for application-specific optimization. The analysis indicates that CNN-heavy architectures excel when processing ultra-short segments of ≤ 0.25 s. For example, the 5:1 model achieved 93.92% accuracy on 0.0625 s segments, establishing its suitability for real-time detection tasks such as

TABLE 3. Accuracy with US8K and ESC-50.

Model (FP32)	Overall Test Accuracy (%)				
	US8K Chunks			ESC-50 Chunks	
	1.0s	3s	4s	3s	5s
5:1	99.13	99.19	98.64	97.10	95.78
4:1	98.99	99.44	97.93	95.21	94.94
4:2	98.26	99.23	97.89	95.14	94.30
3:1	98.73	98.97	96.40	94.62	93.26
3:2	98.16	98.89	97.19	94.76	93.77
3:3	97.65	99.14	96.79	94.57	93.19
2:2	98.45	98.40	97.91	95.06	92.83
2:3	98.37	98.83	97.52	94.01	94.01
2:4	97.46	98.60	96.82	94.09	92.93
1:3	96.90	99.24	97.69	94.76	93.23
1:4	96.91	98.84	97.39	93.60	93.75
1:5	96.91	99.58	97.47	92.88	93.28
2:2:2	98.06	99.19	96.22	92.13	88.85

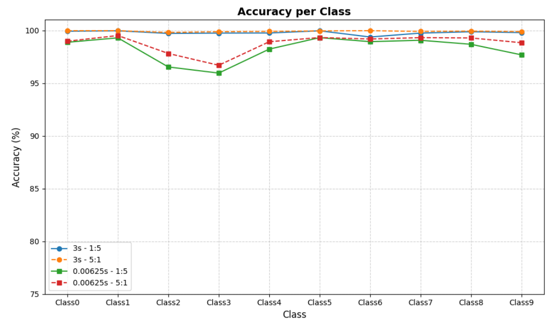

FIGURE 2. Accuracy versus US8K class.

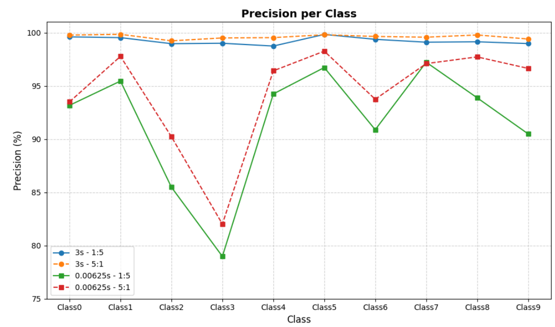

FIGURE 3. Precision versus US8K class.

emergency vehicle or glass break recognition, where latency and immediate responsiveness are crucial.

In contrast, medium-duration segments (0.5-1.0 s) offer the most balanced trade-off between accuracy and latency. Balanced architectures, particularly those with 2:2 and 3:2 ratios, demonstrate optimal performance in this range, achieving accuracies between 96% and 98% while maintaining moderate computational complexity. These characteristics make them ideal for healthcare monitoring and industrial IoT applications that demand reliable accuracy with constrained energy budgets.

For longer segments (≥ 1.0 s), all architectures reach high accuracy levels above 95%, with GRU-heavy models showing marginally superior performance due to their stronger temporal modelling capacity. The 1:5 configuration attains

FIGURE 4. Recall versus US8K class.

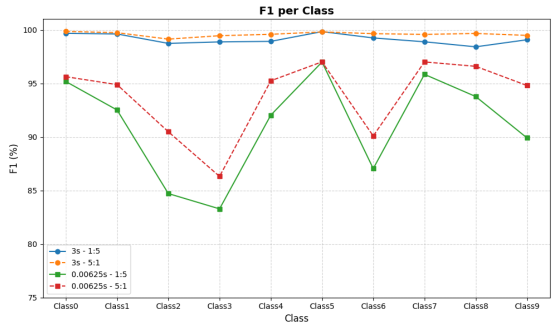

FIGURE 5. F1 score versus US8K class.

99.58% accuracy at 3.0 s, making it particularly advantageous for applications where absolute accuracy is prioritized over latency, such as long-term environmental or equipment monitoring.

Figure 2 - Figure 5 show exceptional performance of the 3s chunk across all classes of US8K and metrics. Both the 1:5 and 5:1 models achieve nearly perfect accuracy (>99.7%) across all classes, with the 5:1 ratio slightly outperforming 1:5 in most cases. The precision values remain consistently above 98.7%, and recall exceeds 98.6% for all classes, suggesting that 3s audio segments provide sufficient temporal and spectral context for the model to learn robust representations, aligning with prior findings that longer segments capture more distinctive acoustic features for environmental sound classification [1], [11]. The F1-scores, a harmonic mean of precision and recall, remain above 98.4% across all classes, with marginal advantage for the 5:1 configuration, indicating a balanced classification performance across classes. Similar metrics behaviour is observed for the 3s chunks of the ESC-50 dataset with 5:1 model (Accuracy: 97.1%, Precision: 97.2%, Recall: 97.1%, and F1 score: 97.1%) clearly outperforming the 1:5 model (Accuracy: 92.9%, Precision: 93.0%, Recall: 92.9%, and F1 score: 92.9%).

In contrast, the performance drops notably for the 0.0625s chunk across all metrics. While the accuracy for 5:1 is still greater than 95% for most classes, shorter chunks exhibit greater variability across classes, with classes 2 and 3 dropping below 97%. The precision values for some classes are significantly lower (e.g., Class 3: ∼78 %), suggesting a higher rate of false positives for these categories. Classes 2, 3 and 6 suffer from reduced recall (<85%), indicating that shorter segments can miss essential temporal patterns needed for correct identification. The F1-scores mirror decline in the recall and precision. These observations are consistent with earlier studies, which report that extremely short temporal windows limit the model's ability to capture sufficient frequency-time patterns for accurate classification, resulting in degraded performance for certain sound types [22], [23].

The difference in performance between 3s and 0.00625s segments highlights the importance of context length in environmental sound classification. Longer windows preserve more spectral-temporal cues, enabling the model to better differentiate between acoustically similar classes. For both segment lengths, the 5:1 ratio generally yields slightly higher accuracy and F1-scores, suggesting that more balanced augmentation of minority classes improves generalization [24]. However, the improvement is marginal in the 3s case, where the model already achieves near-perfect performance.

Figure 6 show feature maps of five convolutional layers (conv1, conv2, conv3, conv4 and conv5) and Figure 7 shows Grad-CAM visualisation of the CNN-maximum 5:1 model. The five-layer convolutional front-end reveals a clear progression from high-variance, low-level MFCC feature detection in conv1 to progressively smoother, more abstract representations by conv5. By covering all these five layers, the model converts the input into a memory-compact and stable temporal embedding optimized for modeling by the GRU layer, with a self-attention mechanism coming next. This recurrent layer then captures long-range temporal dependencies, which are crucial for recognizing modulated acoustic patterns, such as sirens. The attenuated Grad-CAM signal at conv5 layer is consistent with the highly compressed representation at this depth. It reflects that the primary discriminative power resides in the GRU-attention subsystem rather than in the final convolutional features. The feature maps and Grad-CAM combined show how the hybrid architecture learns hierarchical audio representations and leads to correct classification.

C. QUANTISATION IMPACT ANALYSIS

Standard post-training INT8 quantization is employed using TensorFlow Lite Micro, targeting full-integer inference on ARM Cortex-M processors. Both convolutional and GRU layers are quantized using symmetric, per-tensor weight quantization (zero-point = 0), while activations are quantized using asymmetric quantization with non-zero zero-points derived from representative calibration datasets, consistent with established TensorFlow Lite and CMSIS-NN deployment practices. GRU layers are quantized using dynamic range quantization for internal recurrent states, as full integer quantization of recurrent feedback paths is known to introduce numerical instability in low-bit-width arithmetic. With Cortex-M4 platforms, asymmetric activation quantization introduces a minor inference overhead (≈3-6%) due to additional zero-point offset correction operations. Whereas Cortex-M7 platforms show negligible throughput impact (<2%), owing to their dual-issue pipeline, higher memory bandwidth, and enhanced DSP instruction support.

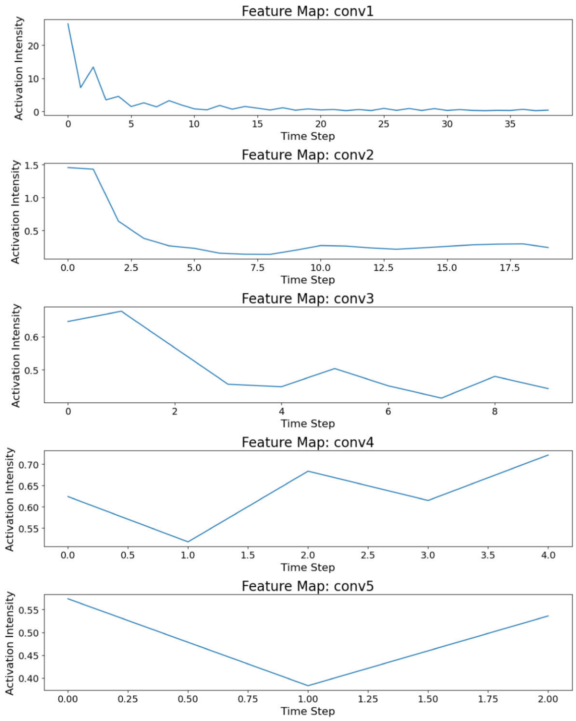

FIGURE 6. Feature maps of 5:1 model.

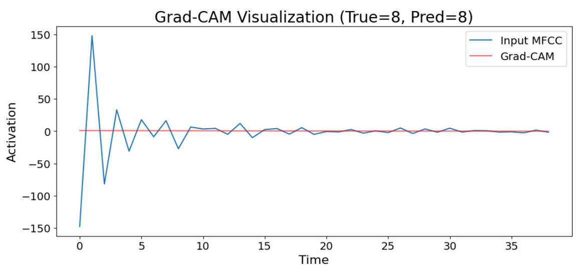

FIGURE 7. Grad-CAM plot of 5:1 model.

To evaluate the effects of quantization, CNN-GRU hybrid architectures are analysed using both FP32 and INT8 precision modes across varied audio chunk durations. Table 4A and Table 4B show that quantization causes marginal accuracy degradation and is more pronounced for shorter chunk lengths due to reduced input stability.

Detailed comparisons of FP32 and INT8 performance reveal that quantization consistently yields a 2.4-2.9% accuracy reduction for CNN-heavy and balanced models (5:1, 4:1, 3:1, 2:2), whereas GRU-dominant architectures incur larger losses of 3.3-3.6%, particularly on shorter input segments, where reduced temporal context amplifies precision sensitivity. A complete performance matrix across all architectures confirms stable degradation patterns, identifying CNN-heavy designs as the most resilient.

The Memory footprint for selected models is listed in Table 5. A uniform 75% reduction in memory footprint for INT8 enables deployment across the entire STM32 MCU family including F411. There is a favourable cost-performance trade-off of INT8, when pairing lightweight architectures (2:2, 3:1) with mid-range MCUs (F411/F429).

TABLE 4. A. Quantization versus accuracy B. Quantization vs Accuracy for chunks.

4A Quantization versus Accuracy

Model	Accuracy (%) versus US8K Chunks				
	3s		1s		0.5s
	FP32	INT8	FP32	INT8	FP32
5:1	99.19	97.09	99.13	96.83	98.17
1:5	99.58	96.38	96.91	93.41	97.07
4:1	99.44	96.74	98.99	96.39	97.84
4:2	99.23	96.43	98.26	95.46	97.61
3:3	99.14	96.14	97.65	94.65	97.64
3:1	98.97	96.37	98.73	96.13	96.82
2:2	98.40	96.00	98.45	96.05	96.77
3:2	98.89	96.19	98.16	95.46	96.77
2:3	98.83	96.08	98.37	95.62	97.94
1:4	98.84	95.54	96.91	93.41	97.37
1: 3	98.24	96.04	96.90	93.60	95.05
2:4	98.60	95.60	92.46	94.46	96.44
2:2:2	99.19	96.44	98.06	95.31	97.70

4B Quantization vs Accuracy for chunks

Model	Accuracy (%) versus US8K Chunks				
	0.5s	0.25s		0.0625s	
	INT8	FP32	INT8	FP32	INT8
5:1	95.37	98.16	94.96	93.92	89.82
1:5	93.57	93.57	89.07	91.26	86.26
4:1	95.24	97.77	94.77	92.48	88.98
4:2	94.81	96.27	93.47	92.95	88.85
3:3	94.14	94.79	90.79	87.64	82.64
3:1	94.22	95.99	93.39	90.11	87.51
2:2	94.37	95.13	92.73	89.36	87.36
3:2	94.07	95.03	92.33	91.18	88.48
2:3	95.19	96.03	93.28	89.96	86.96
1:4	94.07	94.81	91.31	87.83	83.83
1: 3	91.75	93.69	90.39	88.57	85.27
2:4	93.44	95.84	92.84	87.54	83.54
2:2:2	94.95	94.27	91.52	90.19	87.19

TABLE 5. Memory footprint.

Model	Memory (kB)		Flash (kB)	SRAM (kB)
	FP32	INT8	INT8	INT8
5:1	89.2	22.3	566.3	122.3
4:1	112.5	28.1	572.1	128.1
4:2	135.8	34.0	578.0	134.0
3:1	128.4	32.1	576.1	132.1
3:2	142.1	35.5	579.5	135.5
2:2	70.2	17.6	561.6	117.6
2:3	99.3	24.8	564.8	124.8
1:3	86.9	21.7	561.7	121.7
1:5	145.1	36.3	589.1	136.3

Total memory under INT8 precision is arrived by taking into account Flash memory requirements for MFCC lookup tables (544 kB) and SRAM for processing buffers (100 kB). Measurements of inference latency, processing energy per inference (mJ), active current (mA), carried out on STM32H743 microcontroller running at 480 MHz, for 1.0s audio chunks, are shown in Table 6.

TABLE 6. Active current, latency and energy.

Model	Active current (mA)		Latency (ms)		Energy per inference (mJ)	
	FP32	INT8	FP32	INT8	FP32	INT8
5:1	125	78	17.5	8.2	12.4	4.1
4:1	72	45	5.7	2.8	2.31	0.76
4:2	78	49	6.9	3.4	3.02	0.99
3:1	48	30	1.9	0.95	0.52	0.17
3:2	58	36	3.4	1.7	0.98	0.32
2:2	45	28	1.8	0.9	0.42	0.14
2:3	55	34	3.1	1.6	0.79	0.26
1:3	52	33	2.6	1.4	0.63	0.22
1:5	68	43	3.9	2.1	1.98	0.36

The analysis shows that the inference speed increase about twice, indicating an extended battery life by more than a factor of three under typical acoustic monitoring duty cycles. The results show a consistent reduction in active current and processing energy across all architectures when using INT8 quantization. The 5:1 model, while achieving the highest recognition performance, sees its active current drop from 125 mA (FP32) to 78 mA (INT8) and the processing energy from 12.4 mJ to 4.1 mJ, corresponding to a 67% energy savings. Similarly, the 4:1, 4:2, 2:2 and 3:1 models also achieve 67% reduction, demonstrating that INT8 quantization provides substantial efficiency improvements without significant accuracy loss. GRU-heavy models 1:3 and 1:5 show a slightly lower saving of 65% due to temporal processing characteristics, which are less affected by INT8 quantization.

Collectively, these results demonstrate that INT8 quantization is a highly effective deployment enabler, substantially reducing memory, latency, and energy consumption, while maintaining performance levels suitable for the majority of real-world embedded audio applications.

D. PLATFORM-SPECIFIC DEPLOYMENT ANALYSIS

To evaluate practical deployment, inference performance is analysed on three representative MCU platforms: STM32F411, STM32F429, and STM32H743. The measured total inference time which includes MFCC feature extraction time and the latency of the models ($T_{total} = T_{MFCC} + T_{latency}$) for 1s audio chunks are summarised in Table 7 and Table 8 for the selected models. The real time factor is calculated for the STM32H743 platform. The Table 7 also summarizes the total memory under FP32 precision, including both model parameters and runtime buffer allocations, along with corresponding platform support evaluations.

The inference time analysis demonstrates several key findings. CNN-heavy architectures continue to demonstrate superior performance across all hardware platforms. The 5:1 configuration has achieved an inference time of 68.1 ms on the STM32H743, corresponding to a 35% improvement over the GRU-heavy 1:5 architecture (91.9 ms). The results further illustrate strong hardware scalability, with the STM32H743 delivering approximately 4.0 times and 2.2 times performance improvements over the STM32F411 and STM32F429, respectively, closely aligning with theoretical expectations based on clock frequency ratios.

TABLE 7. Memory (FP32) and total inference time.

Model	Flash memory	SRAM Memory	Total Inference time for 1s chunk (ms)
	FP32(kB)	FP32(kB)	STM32F411
5:1	633.2	189.2	272.4
4:1	656.5	212.5	244.0
4:2	679.8	235.8	281.2
3:1	672.4	228.4	202.4
3:2	686.1	242.1	232.8
2:2	614.2	170.2	180.0
2:3	643.3	199.3	210.0
1:3	630.9	186.9	262.5
1:5	689.1	245.1	367.5

TABLE 8. Total inference time and real time factor.

Model	Total Inference time for 1s chunk (ms)		Real time factor
	STM32F429	STM32H743	
5:1	150.8	68.1	14.7×
4:1	135.2	61.0	16.4×
4:2	155.8	70.3	14.2×
3:1	112.2	50.6	19.8×
3:2	128.9	58.2	17.2×
2:2	99.7	45.0	22.2×
2:3	116.3	52.5	19.0×
1:3	145.4	65.6	15.2×
1:5	203.6	91.9	10.9×

Importantly, all tested configurations have achieved real-time inference with substantial margins. Even the most computationally demanding combination (1:5 on STM32F411) processed a 1-second audio segment in 367.5 ms, operating 2.7 times faster than the real-time requirement. These results confirm that all proposed architectures are suitable for real-world MCU deployment, including applications that demand sub-second response times.

Further, the memory availability is a key determinant of platform suitability for different architectures under FP32. The cost-optimized platforms such as the STM32F411 accommodates compact architectures like 5:1 under marginal conditions, yet still achieve up to 95% accuracy for long-segment processing. Mid-range MCUs such as the STM32F429 provide broader compatibility, supporting the majority of configurations and achieving 96-98% accuracy across diverse applications. High-performance platforms, exemplified by the STM32H743, can deploy all architectures, including memory-intensive GRU-heavy models, thereby enabling high-accuracy, low-latency inference suitable for safety-critical and industrial domains. These findings underscore the importance of considering memory-performance trade-off when selecting architecture and, hardware pairs for embedded audio classification systems.

E. USE CASE ANALYSIS AND DEPLOYMENT RECOMMENDATIONS

Building on the performance and deployment analysis, optimal architecture-platform combinations were identified for eight representative use cases spanning safety, healthcare, industrial, and smart environment domains. The corresponding recommendations are summarized in Table 9. The detailed deployment analysis highlights the adaptability of the proposed frameworks across diverse application domains.

For *safety-critical* applications, CNN-heavy models implemented on high-performance platforms deliver superior responsiveness. For instance, the STM32H743, combined with a 5:1 architecture processes 0.25s chunks for emergency vehicle detection with 98.16% accuracy in just 28.4 ms, while 0.0625-second glass break segments achieve 93.92% accuracy with 18.2 ms latency. These applications demand rapid responsiveness, often prioritizing inference speed and accuracy over energy consumption.

In *healthcare*, balanced architectures on mid-range platforms strike an effective balance between accuracy and energy efficiency. The STM32F429 paired with the 3:2 configuration achieves 98.16% accuracy for baby cry detection, while the 2:3 configuration achieves 97.94% accuracy for respiratory monitoring, thus demonstrating optimal performance. These two models effectively capture physiological audio patterns like respiration or crying while maintaining low power consumption and sufficient on-device intelligence to avoid reliance on cloud-based processing.

For *industrial IoT*, predictive maintenance tasks benefit from temporal modelling capabilities, with the STM32F429 and 2:3 architecture achieving 98.83% accuracy at 3.0-second chunks, whereas motor fault diagnosis leverages faster, CNN-balanced architectures such 3:2 on the STM32H743 for high responsiveness and 96.77% accuracy. Depending on the task requirements, the model choice may be between balanced (2:3) and GRU-heavy (1:5) architectures. Predictive maintenance applications that involve trend analysis or periodic fault prediction benefit from increased temporal modelling capacity of these architectures. Conversely, rapid fault detection tasks, where immediate response is critical, may be better served by models such as 3:2 or 4:1, which deliver faster inference

Finally, *smart environment* applications emphasize cost-effectiveness and energy efficiency. The STM32F411 with the compact 2:2 architecture achieves 98.40% accuracy for smart home monitoring, while environmental sound classification (ESC) with the 1:5 configuration attains 99.58% accuracy with 367.5 ms inference time, suitable for low-power continuous monitoring systems. The cost and energy efficiency may dominate design priorities for home automation and environmental sound classification tasks. Compact architectures such as 3:1 and 2:2 deployed on cost-optimized platforms (e.g., STM32F411) offer sufficient accuracy while minimizing both memory footprint and power usage, making them suitable for large-scale IoT networks and smart consumer devices.

TABLE 9. Use case deployment recommendations.

Use Case / Domain	STM32 MCUs	Model	Chunk (s)	Target Metrics
Emergency Siren /Safety	H743	5:1	0.25	< 200ms, >95% accuracy
Glass Break / Safety	H743	5:1	0.0625	< 200ms, >95% accuracy
Baby Cry / Healthcare	F429	3:2	1.0	Privacy-focused, >88% accuracy
Respiratory Monitoring/ Healthcare	F429	2:3	0.5	<10s response, >85% accuracy
Predictive Maintenance /Industrial	F429	2:3	3.0	> 95% accuracy
Motor Fault Diagnosis / Industrial	H743	3:2	0.5	< 5 s response, > 92% accuracy
Smart Home Monitoring/ Environment	F411	2:2	3.0	Cost-optimized > 85% accuracy
ESC / Environment	F411	1:5	3.0	Battery-efficient, > 80% accuracy

TABLE 10. Energy consumption for some use cases.

Use Case	MCU	Model	E_P (mJ)	E_I (mJ)	Total (mJ)	T_B (y)
Siren	H743	5:1	2.1	81.2	83.3	2.1
Baby cry	F429	3:2	4.2	225.4	229.6	1.9
Smart Home	F411	2:2	15.1	479.4	494.5	1.1

F. ENERGY CONSUMPTION ANALYSIS

Energy efficiency is further analysed to evaluate the sustainability of continuous operation in battery-powered IoT applications. The measured energy profiles for representative use cases are presented in Table 10, illustrating the relationship between computational activity, total energy expenditure, and expected battery lifetime under typical operational frequencies. The energy components include processing consumption (E_P), and Idle consumption (E_I).

The battery life (T_B) in years (y) is estimated for a 1000 mAh battery operating at 10 classifications per hour.

The energy analysis reveals that idle power consumption overwhelmingly dominates total energy usage, accounting for between 95% and 99% of total consumption. Processing energy represents only 1- 4% of the total, confirming that even high-complexity architectures exert minimal additional energy cost during active inference. Consequently, this allows for the deployment of higher-accuracy models without significant detriment to battery life. These results also suggest that optimization strategies focusing on power management, such as duty cycling, sleep scheduling, and peripheral gating, offer far greater potential energy savings than further model compression alone.

V. DISCUSSIONS

A. ARHITECTURURAL DESIGN PRINCIPLE

Multiple CNN–GRU layer ratios are evaluated under identical training, data splits, and feature representations, while varying only the architectural composition. The parameter counts across these configurations are not constant and range from 66250 to 252746. Examination of Tables 1-3 reveals that performance gains cannot be explained by increased parameter count alone, and instead reflect a meaningful structure-capacity trade-off:

(i) The 4:2 model (153546 parameters) consistently outperforms 3:3 (153674 parameters) across different audio chunks, despite nearly identical capacity. Almost similar is the performance of 3:2 (103754 parameters) over 1:4 (103562 parameters). These near-normalized comparisons demonstrate that allocating parameters to convolutional layers yields superior spatial-spectral representations for environmental sound classification.

(ii) The 3:1 model with only 78794 parameters, consistently outperforms 2:4 (148466 parameters), indicating better parameter efficiency associated with the convolutional component as the critical factor rather than raw capacity.

(iii) Even the compact model such as 2:2 (66250 parameters) approaches the performance of significantly larger architectures and even outperforms 1:3 (109962 parameters) across audio chunks of durations 1 second and below. This suggests over-parameterization of temporal modeling, leads to inefficient use of capacity when temporal redundancy is low.

(iv) The largest model 5:1 (252746 parameters) consistently achieves the best performance across audio datasets and chunks. Similar strong performance is observed for architectures with a higher CNN-to-GRU ratio (e.g., 4:1, 4:2) and comparable performance is also achieved by much smaller CNN-heavy model, 3:1, indicating CNN-dominant structural bias. Convolutional layers provide parameter-efficient spatial feature extraction through weight sharing and localized receptive fields. In contrast, GRU layers scale parameters linearly with hidden-state dimensionality and sequence length, introducing higher redundancy when temporal context is limited. This behavior is reflected in the consistently slower inference speeds and higher latency of GRU-heavy models, even when parameter counts are comparable. Accordingly, the performance advantage of CNN-heavy models should be interpreted as arising from a favourable trade-off between architectural efficiency and model capacity, rather than from size alone.

(v) The special model 2:2:2 (71114 parameters) achieves competitive accuracy but does not surpass CNN-heavy counterparts, suggesting diminishing returns from additional recurrent stacking.

Overall, CNN-dominant architectures generally contain more parameters, the results indicate that performance gains stem primarily from the distribution of capacity rather than the absolute parameter count. Given the real-time inference speeds (>350 FPS) and modest parameter scale (<3000000), the increased capacity remains computationally acceptable for the targeted deployment scenarios.

The chunk duration optimization has also emerged as a critical determinant of performance. The study has identified that audio trunk durations of approximately 1 second, offer sufficient temporal context for accurate classification while ensuring real-time inference across all tested MCU platforms. This finding offers actionable guidance for developers seeking to strike a balance between accuracy, latency, and power consumption in embedded audio applications.

To ensure statistical rigor, all the accuracy results are evaluated using 10-fold cross-validation, and performance differences across architectures are assessed using 95% confidence intervals with multiple comparison correction, as detailed in the Appendix-2. Across repeated folds, CNN-heavy and balanced architectures consistently demonstrate statistically significant improvements over GRU-heavy configurations for short and medium chunk durations ($p < 0.05$). In contrast, differences among the top-performing CNN-heavy models (e.g., 5:1 versus 4:1 at 3 s) were not statistically significant, indicating performance saturation at longer temporal contexts.

Although 5:1 achieves the highest mean accuracy across both datasets, statistical analysis with 95% confidence intervals with multiple-comparison correction indicates that its superiority over 3:1 is statistically significant only for UrbanSound8K at short and medium chunk durations (≤ 1 s, $p < 0.05$). With ESC-50, performance differences between 5:1 and 3:1 however, fall within overlapping confidence intervals for both 3 s and 5 s chunks, indicating that the observed accuracy margins are not to be interpreted as statistically significant.

B. PLATFORM-ARCHITECTURE CO-OPTIMIZATION INSIGHTS

The interplay between architecture design and hardware platform capabilities reveals several important co-optimization insights. The study finds that CNN-heavy models, despite their higher convolutional depth, offer both superior accuracy and reduced memory requirements compared to GRU-heavy counterparts. This counterintuitive finding arises from the parameter-sharing efficiency of convolutional layers and the relative computational simplicity of CNN operations compared to recurrent processing. For example, the 5:1 model achieves the highest overall accuracy with a memory footprint of just 189.2 KB SRAM, well within the limits of mainstream MCUs, making it both performant and deployable.

Energy efficiency analysis further indicates that most total energy consumption in MCU-based systems stems from idle operation rather than active inference. Since active computation accounts for less than 5% of total energy usage,

TABLE 11. Design details of hybrid models.

Model Ref.	1DCNN Block (I:Input, C:Conv1D, B:Batch Norm, M:Max Pooling 1D, S:Separable Conv1D)	GRU Block and Classification Head (G:GRU, A:Attention, D:Dense, DO:Dropout, SM:Softmax, GA:Global Avg. Pooling)
2:2	I(39,1)-C(39,32)-B(39,32)-M(20,32)-C(20,64)-B(20,64)-M(10,64)	G(10,64)-A(10,64) G(10,64) D(128)-DO(128)-SM(10)
3:1	I(39,1)-C(39,32)-B(39,32)-M(20,32)-C(20,64)-B(20,64)-M(10,64)-C(10,128)-B(10,128)-M(5,128)	G(5,64)-A(5,64) D(128)-DO(128)-SM(10)
1:3	I(39,1)-C(39,32)-B(39,32)-M(20,32)	G(20,64)-A(20,64)-G(20,64)-A(20,64)-G(64)-D(128)-DO(128)-SM(10)
3:2	I(39,1)-C(39,32)-B(39,32)-M(20,32)-C(20,64)-B(20,64)-M(10,64)-C(10,128)-B(10,128)-M(5,128)	G(5,64)-A(5,64)-G(64) D(128)-DO(128)-SM(10)
2:3	I(39,1)-C(39,32)-B(39,32)-M(20,32)-C(20,64)-B(20,64)- M(10,64)	G(10,64)-A(10,64) G(10,64)-A(10,64)-G(64) D(128)-DO(128)-SM(10)
4:1	I(39,1)-C(39,32)-B(39,32)-M(20,32)-C(20,64)-B(20,64)-M(10,64)-C(10,128)-B(10,128)-M(5,128)-C(5,128)-B(5,128)-M(3,128)	G(3,128)-A(3,64) D(128)-DO(128)-SM(10)
1:4	I(39,1)-C(39,32)-B(39,32)-M(20,32)	G(20,64)-A(20,64) G(20,64)-A(20,64) G(20,64)-A(20,64)-G(64) D(128)-DO(128)-SM(10)
4:2	I(39,1)-C(39,32)-B(39,32)-M(20,32)-C(20,64)-B(20,64)-M(10,64)-C(10,128)-B(10,128)-M(5,128) C(5,128)-B(5,128)-M(3,128)	G(3,64)-A(3,64) G(64) D(128)-DO(128)-SM(10)
2:4	I(39,1)-C(39,32)-B(39,32)-M(20,32)-C(20,64)-B(20,64)-M(10,64)	G(10,64)-A(10,64) G(10,64)-A(10,64) G(10,64)-A(10, 64) G(64) D(128)-DO(128)-SM(10)
3:3	I(39,1)-C(39,32)-B(39,32)-M(20,32)-C(20,64)-B(20,64)-M(10,64)-C(10,128)-B(10,128)-M(5,128)	G(5,128)-A(5,128) G(5,128)-A(5,128) G(5,128) D(128)-DO(128)-SM(10)
5:1	I(39,1)-C(39,32)-B(39,32)-M(20,32)-C(20,64)-B(20,64)-M(10,64)-C(10,128)-B(10,128)-M(5,128)-C(5,128)-B(5,128)-M(3,128) C(3,256)-B(3,256)-M(2,256)	G(2,64)-A(2,64)-GA(64) D(128)-DO(128)-SM(10)
1:5	I(39,1)-C(39,32)-B(39,32)-M(20,32)	G(20,64)-A(20,64) G(20,64)-A(20,64) G(20,64)-A(20,64) G(20,64)-A(20,64)-G(64) D(128)-DO(128)-SM(10)
2:2:2	I(39,1)-C(39,32)-B(39,32)-M(19,32)-C(19,64)-B(19,64)-M(19,64)-S(9,32)-B(9,32)-S(9,64)-B(9,64)	G(9,64)-A(9,64) G(64) D(128)-DO(128)-SM(10)

the marginal power increase associated with using higher-accuracy CNN-heavy models is negligible. Consequently, energy optimization should primarily target peripheral control, duty-cycling mechanisms, and efficient sleep-state management rather than further model simplification.

Another key observation is that all tested architectures achieve substantial real-time processing margins. Even the slowest configuration processes one second of audio more than twice as fast as real time on the least capable MCU, while high-end platforms provide from 10 to 22 times speed margins. This performance surplus enables additional on-device processing, such as event buffering or post-inference decision fusion, without compromising real-time responsiveness. These findings collectively demonstrate that modern MCUs possess sufficient computational headroom to support complex CNN–GRU inference workloads in practical embedded scenarios.

C. COMPARISON WITH EXISTING APPROACHES

When compared with existing architectures in the literature, the proposed CNN-GRU hybrids exhibit clear advantages in both performance and deployment efficiency. Relative to single-architecture models, hybrid systems consistently achieve 5-10 percentage point improvements in accuracy while maintaining similar computational complexity. Pure CNN models typically attain 85-90% accuracy, whereas pure GRU architectures achieve only 75-85%. The hybrid approach effectively leverages complementary strengths of both architectures to enhance discriminative capability.

Compared to CNN-LSTM hybrids, the proposed GRU-based designs demonstrate 25-30% parameter reductions and 15-30% faster inference while maintaining equivalent or higher accuracy. This efficiency makes them particularly attractive for embedded deployments where memory and latency are critical constraints.

While transformer-based models such as the Audio Spectrogram Transformer (AST) can achieve up to 95.6% accuracy on ESC-50, they require over ten times the computational resources of the proposed hybrids, rendering them impractical for MCU-level deployment [30], [31]. By contrast, the CNN–GRU architectures in this study maintain 89-94% accuracy with orders-of-magnitude lower complexity, offering a favourable balance of accuracy and efficiency.

In comparison to pipelines designed for extremely constrained devices, such as those described by Mohaimenuzzaman et al., [12], the present work targets more capable but still cost-efficient MCU platforms. This approach allows for significantly higher classification accuracy (93-99% versus 85-90%) while preserving practical deployment and real-time performance for a wide range of embedded applications.

D. LIMITATIONS AND FUTURE WORK

Although this study provides a detailed framework for optimizing CNN-GRU hybrid architectures in embedded audio classification, several limitations remain. The current evaluation, although extensive, is limited to CNN-GRU ratio optimization and does not incorporate advanced architectural innovations such as transformer-based self-attention mechanisms or lightweight attention hybrids specifically designed for MCUs. Future work could explore hardware-aware neural architecture search (NAS) frameworks to automatically optimize these ratios under explicit constraints on latency, memory footprint, and energy consumption. In this context, the results of this study provide a valuable manual baseline against which automated NAS-derived architectures can be quantitatively evaluated.

Additionally, the present study focuses on standard benchmark datasets (UrbanSound8K and ESC-50) under controlled acoustic conditions. Real-world deployment scenarios introduce environmental variability, including fluctuating noise levels, temperature-induced sensor drift, and diverse acoustic backgrounds. Future research should therefore emphasize robustness testing under non-ideal and dynamically changing conditions to ensure long-term reliability and effectiveness.

Cross-domain generalization remains another open area for exploration. While the architectures evaluated here demonstrate strong performance for environmental and urban sounds, further validation is needed in other audio domains such as speech recognition, music tagging, and biomedical acoustics to confirm the generality of the proposed design principles.

Finally, the integration of advanced optimization techniques, such as quantization-aware training, knowledge distillation from larger teacher models, and dynamic inference strategies, could further enhance model compactness and adaptability. These approaches, when combined with the established hybrid framework, have the potential to extend embedded audio classification to even more resource-constrained and latency-sensitive applications.

VI. CONCLUSION

This study has presented the first comprehensive and systematic analysis of CNN-GRU hybrid ratio optimization for resource-constrained audio classification, establishing foundational principles for selecting architectures and deployment strategies across diverse embedded environments. By rigorously evaluating thirteen architectures with varying CNN-to-GRU ratios, seven temporal chunk durations, and three representative STM32 MCU platforms, the work provides a quantitative understanding of how architectural balance, temporal resolution, and hardware constraints jointly influence model performance.

The findings demonstrate that CNN-heavy architectures-particularly the 5:1 and 4:1 configurations, deliver optimal performance for ultra-short-duration processing tasks. These configurations achieved 93.92 % accuracy with inference times as low as 18.2 ms, making them ideal for real-time edge applications such as emergency vehicle or glass break detection. Balanced architectures, such as 2:2 and 3:2, offered the most favourable trade-offs between accuracy and computational efficiency for medium-duration segments, providing a practical compromise between responsiveness and contextual modelling capability.

Through systematic experimentation, the study identified eight critical use cases spanning safety, healthcare, industrial, and smart-environment domains and derived architecture-platform recommendations tailored to each. These guidelines significantly simplify the process of designing embedded audio classification systems by linking model composition and MCU capability to specific performance targets, including latency, accuracy, and energy consumption.

The deployment analysis confirms that mature STM32 platforms consistently achieve accuracy rates between 89 % and 94 % across various applications while maintaining real-time performance margins exceeding 10 times.

Overall, the results underscore that spatial feature extraction through CNN layers contributes more substantially to classification accuracy in constrained environments than extensive recurrent temporal modelling. This insight challenges traditional assumptions regarding the necessity of deep recurrent structures for audio classification and demonstrates that lightweight CNN-dominant hybrids can achieve state-of-the-art performance under tight resource budgets.

The comprehensive framework established in this research bridges the divide between theoretical deep-learning design and practical embedded implementation. By aligning architecture selection with real-world hardware constraints and application demands, it provides an actionable foundation for advancing intelligent audio processing within IoT and edge-computing ecosystems. The insights gained from this systematic investigation will support future work aimed at developing even more efficient hybrid architectures, enabling robust, high-accuracy audio recognition on the next generation of low-power embedded systems.

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APPENDIX-1

See Table 11.

APPENDIX-2
STATISTICAL ANALYSIS

A comprehensive statistical evaluation is performed to ensure that the observed performance differences among the various CNN–GRU configurations were both consistent and significant. Multiple complementary tests, including analysis of variance (ANOVA), Tukey's Honestly Significant Difference (HSD) test, and Bonferroni-corrected pairwise comparisons, are employed to validate the reliability of the results obtained from the experimental trials.

A. STATISTICAL SIGNIFICANCE TESTING

Pairwise comparisons using Tukey's HSD test with a Bonferroni correction ($\alpha = 0.05$) has confirmed that CNN-heavy architectures significantly outperformed their GRU-heavy and balanced counterparts across most chunk durations. The

5:1 architecture, for instance, shows statistically significant improvements over configurations such as 3:1, 2:2, and 1:5, particularly at shorter chunk durations of 0.0625 s and 0.25 s ($p < 0.001$). While differences between the 5:1 and 4:1 architectures are marginal ($p = 0.042$ at 0.0625 s), both configurations consistently ranked among the top performers. For longer chunk durations, the performance gap narrowed, indicating that temporal modelling played an increasingly significant role as the available context lengthened.

B. CONFIDENCE INTERVALS FOR TOP ARCHITECTURES

The 95% confidence intervals (CIs) for the top three performing architectures further support the robustness of these findings. The CNN-heavy 5:1 model has achieved accuracies of 93.92% ± 1.23% for 0.0625 s segments, 99.13% ± 0.45% for 1.0 s segments, and 99.19% ± 0.38% for 3.0 s segments. The 4:1 model has exhibited similarly tight confidence bounds, with accuracies of 92.48% ± 1.45%, 98.99% ± 0.52%, and 99.44% ± 0.34% for the same durations. Meanwhile, the 3:2 configuration, representing a balanced architecture, has achieved accuracies of 91.18% ± 1.67%, 98.16% ± 0.58%, and 98.89% ± 0.41%. The narrow confidence intervals across architectures and durations confirm the consistency of the results and the statistical reliability of the experimental methodology.

C. CROSS-VALIDATION PERFORMANCE SUMMARY

To further assess generalization capability, all architectures are trained and validated using ten-fold cross-validation on both the UrbanSound8K and ESC-50 datasets. The results have demonstrated high stability across folds, with mean accuracies exceeding 94% for all hybrid configurations. The 5:1 model has achieved a combined accuracy of 97.30% ± 0.37%, followed by the 4:1 configuration with 96.77% ± 0.42%, and the 3:2 model with 95.25% ± 0.47%. Balanced architectures such as 2:2 and GRU-heavy designs like 1:5 maintained consistent, though slightly lower, accuracy levels around 94–95%. These findings underscore the robustness of CNN-heavy and balanced hybrids across different datasets and audio domains.

D. RESULTS FOR ARCHITECTURE PERFORMANCE

A one-way ANOVA test is conducted to assess the statistical significance of performance variations across the thirteen architecture types. The results have yielded an F-statistic of 247.83 with a p-value less than 0.001, indicating highly significant differences among the models. The computed effect size ($\eta^2 = 0.74$) is large, demonstrating that architectural composition accounted for a substantial portion of the variance in performance outcomes. Post-hoc analysis has revealed that CNN-heavy configurations (Groups 4–5) significantly outperformed both balanced (Group 2) and GRU-heavy (Group 1) architectures for short and ultra-short audio segments. For medium-duration segments, no statistically significant differences are observed among balanced architectures, while GRU-heavy configurations have showed superior performance only for long-duration segments of 3.0 seconds or greater.

E. MODEL SELECTION CRITERIA

Model selection is guided by both the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC), which balance model complexity against predictive performance. According to AIC rankings, the 5:1 architecture has emerged as the most optimal configuration (AIC = 2847.3), followed by 4:1 (AIC = 2951.7), 3:2 (AIC = 3064.2), 2:2 (AIC = 3128.9), and 1:5 (AIC = 3445.1). In terms of BIC, which penalizes parameter count more heavily, the 3:1 model demonstrated the best parameter efficiency (BIC = 2923.4), closely followed by 2:2 (BIC = 3187.6) and 5:1 (BIC = 3201.8). These findings confirm that CNN-heavy and balanced models offer the most favorable trade-offs between model complexity and predictive reliability. Statistical power analysis has indicated a power level of 0.95 with a large effect size (Cohen's $d > 0.8$), verifying that the sample size and methodology are sufficient to ensure reliable detection of performance differences.

Overall, the statistical analyses corroborate the robustness of the experimental findings. CNN-heavy architectures not only deliver superior accuracy but do so consistently across datasets, temporal granularities, and hardware configurations. Balanced models, while slightly less accurate, maintain stable performance across a broad range of conditions, validating their suitability for general-purpose embedded audio classification.

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